

From Techno-Economic Surges to Substrate Selection:

A Proposal for Quantifying Technological Transition Dynamics

Abstract

Carlota Perez's theory of technological revolutions provides one of the most powerful frameworks for understanding long-wave economic development. Her model explains how technological revolutions emerge, how financial capital accelerates their installation, how institutional crises generate turning points, and how deployment periods ultimately diffuse new techno-economic paradigms throughout society.

This paper proposes a complementary framework rather than an alternative one.

While Perez explains the recurring structure of technological revolutions, the present framework seeks to explain a narrower question:

Why do certain technological architectures become civilization-scale substrates while others remain niches, regardless of their apparent technical superiority?

To address this problem, I introduce a constraint-removal model that treats technologies as competing mechanisms for expanding reachable state spaces under physical, economic, and institutional constraints.

The central claim is that technological transitions exhibit recurring structural properties independent of domain, and that these properties can be measured quantitatively.

1. The Problem

Perez's framework successfully explains the emergence and diffusion of technological revolutions.

However, within every technological revolution multiple competing architectures exist simultaneously.

Examples include:

- Vacuum tubes versus transistors
- Merchant sailing ships versus steamships
- Railroads versus canal systems
- Proprietary networks versus TCP/IP
- Film photography versus digital sensors
- Mainframes versus personal computers

The question addressed here is:

What determines which architecture ultimately becomes the dominant substrate?

Historical observation suggests that superior performance alone is insufficient.

Many technically impressive systems fail to achieve widespread adoption, while other systems become foundational infrastructure.

The explanation appears to involve a recurring process of constraint removal.

2. The Substrate Selection Hypothesis

The framework proposes that civilization repeatedly selects technologies that remove the greatest number of active constraints while preserving operational stability.

Technologies therefore compete not merely on performance but on their ability to expand the reachable action space of other actors.

A technology becomes a substrate when external systems reorganize around it.

This implies that technological evolution can be viewed as a sequence of transitions:

Constraint-Trapped Frontier

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Expansion State

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Substrate State

↓

Mature Regime

↓

Extraction Spiral



Obsolescence

The hypothesis is that this sequence recurs across otherwise unrelated domains.

3. Operational Variables

The framework currently models four dimensions.

Reachability (Γ_2)

The number of accessible future states enabled by a technology.

Examples:

- Railroads expanded geographic reachability.
- TCP/IP expanded communication reachability.
- Cloud computing expanded computational reachability.

Invariance (I)

The stability of system performance across environments and use cases.

Examples:

- Electrical grids exhibit high invariance.
- Early aviation exhibited low invariance.
- Modern semiconductors exhibit high invariance.

Velocity (dl/dt)

The rate at which constraints are being removed.

This measures whether a technology is actively improving or stagnating.

Density (D)

The degree to which external actors have reorganized around the technology.

Examples:

- Gas stations around automobiles.
- Software ecosystems around operating systems.
- Internet services around TCP/IP.

Density is especially important because it operationalizes a key insight found throughout Perez's work:

Technologies do not diffuse alone; institutions, capital, and complementary systems reorganize around them.

4. Relationship to Perez's Framework

The proposed framework does not attempt to replace the theory of technological revolutions.

Instead, it may operate at a different analytical layer.

Perez explains:

- why surges emerge,
- why bubbles form,
- why institutional crises occur,
- and how paradigms diffuse.

The present model attempts to explain:

- which architectures survive inside a surge,
- which become infrastructure,
- and which remain trapped as niches.

In this sense, the framework can be viewed as a substrate-selection layer operating within Perez's broader techno-economic surge model.

5. Preliminary Findings

Historical analysis suggests that successful substrates repeatedly display the same structural transition pattern:

1. Initial constraint-trapped frontier.
2. Rapid reduction in friction.
3. External ecosystem formation.
4. Infrastructure lock-in.

5. Maturation.
6. Eventual displacement by a superior constraint-removal mechanism.

Examples include:

- Steam replacing sail.
- Electricity replacing gas lighting.
- Transistors replacing vacuum tubes.
- TCP/IP replacing proprietary networking architectures.
- Digital imaging replacing film.

The recurrence of this pattern suggests the existence of deeper selection dynamics independent of specific technologies.

6. Future Research

The central challenge is falsification.

The framework generates several testable claims:

- Technologies with increasing reachability and increasing density should exhibit higher probabilities of substrate formation.
- Technologies with declining velocity should fail to transition regardless of initial enthusiasm.
- Institutional adoption should lag technical feasibility but precede substrate lock-in.
- Similar transition geometries should appear across unrelated technological domains.

The goal is not to predict specific inventions but to identify the structural conditions under which technological transitions become self-reinforcing.

Conclusion

Perez's work demonstrates that technological revolutions unfold through recurring developmental surges shaped by finance, institutions, and technological change.

This paper proposes that beneath those surges lies another recurring process:

the selection of civilization-scale substrates.

If technological revolutions explain the waves of development, substrate selection may help explain which architectures ultimately survive those waves and become the foundations upon which subsequent economic and social systems are built.

I would be deeply interested in your thoughts on whether this framework complements, contradicts, or extends the techno-economic paradigm approach developed in your work.